Ruben IFRAH

M.Sc. Student in AI – École Polytechnique & Dauphine-PSL

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Education

Dauphine - PSL — 2025 – Present
M2 IASD (AI, Systems, Data) — *Paris, France*

- Highly selective research-oriented Master's program
- Directed by Olivier Cappé & Francis Bach
- Co-accredited ENS-Ulm & Mines Paris

École Polytechnique — 2021 – 2025
Ingénieur Polytechnicien diploma — *Palaiseau, France*

- Majored in Applied Maths and Statistics
- Economics Minor: econometrics, financial models, micro/macro economy
- Strong scientific background: general physics, quantum physics, fluid mechanics, solid mechanics, econo-physics
- GPA: 3.7/4.0

Lycée Saint-Louis — 2019 – 2021
*CPGE PCSI-PSI** — *Paris, France*

- A two-year intensive program in physics, mathematics and engineering sciences
- Top of the 2020-2021 PSI* class

Experience

Clevaligner — June 2025 – Aug. 2025
3D Vision Scientist Intern — *Paris, France*

- Worked in a startup on AI applied to the medical field (dental correction).
- Responsible for assessing and implementing SOTA models in point-cloud teeth segmentation (Transformer-based PointNet).
- Assisted AI engineers in designing and constructing point-cloud pipelines.

University of Valencia — April 2024 – July 2024
Research Assistant (Statistics) — *Valencia, Spain*

- Worked on ecological inference estimators and statistical modeling.
- Developed an extended model using different norms for estimator efficiency.

MBDA — May 2023 – Aug. 2023
Applied Research Intern — *Plessis-Robinson, France*

- Signal Processing & Optimization
- Designed anti-jamming filters for radar systems dealing with stochastic noise.
- Modeled physical constraints using MATLAB simulations to validate robustness.

French Army (Saint-Cyr) — Sept. 2021 – April 2022
Officer Cadet / Platoon Leader — *Martinique & Coëtquidan*

- **Platoon Leader (Jan 2022 - April 2022):** Led a platoon of 30 new recruits soldiers in Martinique. Organized training exercises and managed logistics.
- **Military Academy (Sept 2021 - Jan 2022):** Intensive officer training at Saint-Cyr Coëtquidan. Developed leadership, rigor, and team management skills.

Objective

I am an **Ingénieur Polytechnicien** (X21) currently pursuing the **M2 IASD** (AI, Systems, Data) at Dauphine-PSL. With a background in Physics and **Applied Mathematics**, I am seeking opportunities in **Deep Learning**, with a strong interest in Deep-RL and Computer Vision.

Projects

DPO alignment for Mistral-7B

- Reproduction of Rafailov et al on Anthropic-HH dataset. Achieved 58% win-rate vs SFT baseline. Analyzed the impact of beta parameter on reward collapse

Deep RL Autonomous Racing Agent

- Designed a custom LIDAR-based environment with Procedural Generation (Harmonic Noise) to enforce policy generalization and prevent overfitting. Implemented Dueling DDQN from scratch, demonstrating superior stability by decoupling State and Values streams.

Bayesian Probabilistic Matrix Factorization

- Reproduced Salakhutdinov & Mnih (2008) by implementing a Hierarchical Bayesian model via Gibbs Sampling. Outperformed deterministic baselines by reducing Test RMSE from 0.94 to 0.837

Adversarial Defense via Non-Differentiable Layers

- Developed a robust defense against PGD L_∞ attacks by integrating a Canny Edge filter. Achieved 80% robustness on white-box attacks.

Skills

Deep Learning & AI:

- PyTorch, Gymnasium, Hugging Face, WandB, Scikit-learn

Dev & Scientific:

- Git, Docker, Linux (Bash), LaTeX, MATLAB, R, SQL

Languages:

- French (Native), English (C1 - Fluent), Spanish (B1)

Selected Coursework (M2 IASD)

Learning Theory & Optimization

- Foundations of Machine Learning (*F. Bach*) **16/20**
- Math of Deep Learning (*B. Loureiro*)
- Non-Convex Inverse Problems (*I. Waldspurger*)
- Optimization for ML (*C. Royer*)

Reasoning, RL & GenAI

- Reinforcement Learning (*O. Cappé*) **15/20**
- Monte Carlo Search (*T. Cazenave*)
- Large Language Models (*A. Allauzen*) **17.5/20**
- LLM for Code & Proofs (*M. Lelarge*)

Geometric DL & Computer Vision

- Manifold Learning (*E. Aamari*)
- Point Clouds & 3D Modeling (*J.E. Deschaud*)
- Deep Learning for Image Analysis (*T. Walter*) **18/20**

Scientific Background

- Physics: Quantum Physics, Statistical Physics, Fluid Mechanics
- Maths: Stochastic Calculus, Numerical Analysis, Game-Theory, Differential Equations